

Spontaneous Regression of a Large Vestibular Schwannoma: Is Nonoperative Management Reasonable?

Kok Ann Colin Teo¹ , Rachit Agrwal² , Pin Lin Kei³ , Su Lone Lim³, Siyang Ira Sun³, Shiong Wen Low³ 

¹Division of Neurosurgery, National University Hospital, Singapore

²Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore

³Division of Neurosurgery, Ng Teng Fong General Hospital, Singapore

Received February 21, 2024

Accepted March 25, 2024

Correspondence

Rachit Agrwal

Lee Kong Chian School of Medicine,
Nanyang Technological University,
Clinical Sciences Building (Novena) 11
Mandalay Road, Singapore 308232

Tel: +65-8499-6058

E-mail: rachit004@e.ntu.edu.sg

Vestibular schwannomas (VSs) are the most common cerebellopontine tumors. The natural history of smaller-sized VSs (<30 mm) has been well-studied, leading to the recommendation of a “watch and wait” approach. However, large VSs (>30 mm) have not been extensively studied, mainly because of their rarity. As such, most patients are conventionally offered surgery which carries a significant risk of neurological morbidity. Here, we report a case of a giant VS (>40 mm) in a 30-year-old man who regressed spontaneously. He was lost to follow-up for 18 years and, upon re-presentation, the symptomatology drastically improved and repeat imaging demonstrated a marked reduction in tumor size. Referring to similar cases in other studies, we postulate that most large and giant VSs undergo a phase of growth and stasis, followed by regression due to shifts in the balance between tumorigenic and regressive factors. Taken together with emerging molecular data, further studies are required to better understand the history of large and giant VSs to shape more personalized treatment options. This potentially includes non-operative management as a tenable option.

Keywords Acoustic neuroma; Conservative; Cerebellopontine angle tumors; Remission.

INTRODUCTION

Vestibular schwannoma (VS) is the most common cerebellopontine angle (CPA) tumor in adults, accounting for 8% of all intracranial tumors and more than 80% of all CPA tumors [1]. VS is classically thought to be benign, slow growing, and arises from myelinating Schwann cells of the both superior and inferior vestibular nerves [2,3]. The management of VS is complex and multi-faceted, varying largely between practitioners including in the decision to treat, choice of treatment modality, namely surgery (with variation in approaches including retro sigmoid, trans-labyrinthine, and middle fossa approaches) and radiation (stereotactic radiosurgery [SRS] or stereotactic radiotherapy) [4]. Today, there is no clear evidence supporting one treatment modality over another [5]. While patients should be offered a balanced view of the benefits and risks of

each treatment option, this can be heavily subjected to each individual physician's bias [6].

The natural history of the disease is central to the discussion of the treatment options. Overall, the incidence of VS has been increasing from 0.72 per 100,000 person-years from 1995–1999 to 1.29 per 100,000 person-years from 2012–2016 [7]. Out of all the VS cases, 5% of them occur bilaterally, being associated with neurofibromatosis type 2 (NF2) [8]. However, the natural history of large (30–40 mm) and giant (>40 mm) sporadic unilateral tumors is elusive, given that they are uncommon, constituting 6% of newly diagnosed VS (equivalent to an incidence of 1 VS per million population per year) [9]. Even though these larger tumors were common in earlier studies, another factor to them not being studied well lies in the fact that a large proportion was operated on immediately [10]. This is contrasted to sporadic smaller-sized VS (<30 mm), for which more recent longitudinal studies have provided clarity on their natural history. Most of smaller-sized VS are slow growing, remain static in size, or are even known to regress since diagnosis, which has resulted in a shift of treatment recommendations heavily toward a “watch-and-wait” approach. We

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

Copyright © 2024 The Korean Brain Tumor Society, The Korean Society for Neuro-Oncology, and The Korean Society for Pediatric Neuro-Oncology

report a case of giant sporadic VS that underwent spontaneous regression after 18 years of follow-up and summarize similar observations reported in recent years. We also discuss its implications for the management of large and giant VS, specifically, whether nonoperative management is a reasonable management option for most of this patient population.

CASE REPORT

We present the case of a patient diagnosed with giant left VS (>40 mm) at 30 years of age who presented with episodic vertigo, left-sided hearing loss, and mild gait ataxia. MRI performed at another institution showed a giant VS (Fig. 1). The patient was lost to follow-up for 18 years. He subsequently presented again when he sought medical clearance for employment. Upon review, his vertigo episodes and ataxia were completely resolved, although he no longer had serviceable hearing on his left side. The clinical examination results were otherwise

unremarkable for other neurological deficits. On repeat imaging, there was a >50% reduction in the overall tumor volume, with a significant reduction in the size of both the extrameatal and intrameatal components, along with a mass effect on the brainstem (Fig. 2). No hydrocephalus was observed. The patient was counseled and agreed to undergo regular follow-up for the tumor.

DISCUSSION

Presentation of patients with large VS

Patients with large or giant VS can present in two broad categories: 1) patients with debilitating clinical symptoms and signs attributable to brainstem compression (hearing loss, vertigo, nystagmus, cerebellar signs, lower cranial nerve palsies, or long tract signs), and/or intracranial hypertension with or without obstructive hydrocephalus (e.g., headache, vomiting, blurring of vision, papilledema), “significant” radiological brain-



Fig. 1. Giant vestibular schwannomas (VS) diagnosed at 30 years of age. A: Axial T2 MRI of giant VS. B: Axial T1 MRI of giant VS. C: Coronal T1 MRI of giant VS.

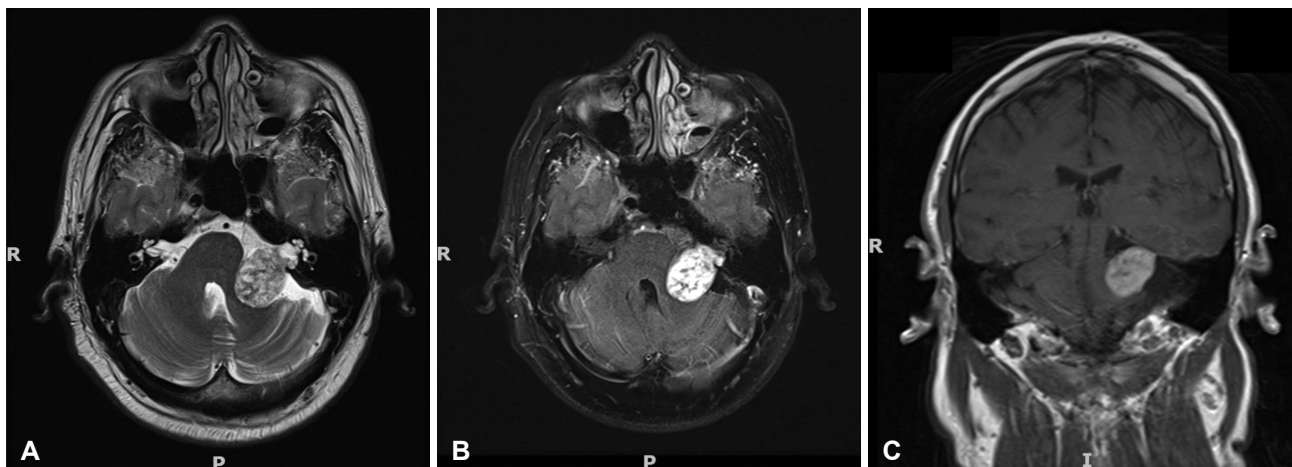


Fig. 2. Marked reduction in tumor volume at 48 years of age. A: Axial T2 MRI of giant VS. B: Axial T2 FLAIR MRI of giant VS. C: Coronal T1 MRI of giant VS. FLAIR, fluid attenuated inversion recovery; VS, vestibular schwannoma.

stem compression and distortion of the fourth ventricle, and rarely when the patient presents in extremis after a tumor bleed; and 2) patients with minimal or no symptoms (“non-debilitating”). While the indications for surgical treatment are clear in the former group, the “watch-and-wait” approach is traditionally not recommended for the latter because of the possibility of life-threatening complications from tumor growth. In a large series of patients with VS who underwent surgical removal by Samii et al. [11], approximately a quarter (26%) of the patients with a large VS (defined as >40 mm) had clinical evidence of intracranial hypertension. Compared with smaller VS (<40 mm), the significant differences in the proportion of patients with neurological deficits were only found in gait ataxia, long tract signs, and lower cranial nerve dysfunction. Notably, the proportion of patients with the latter two neurological issues in the large VS group was low (approximately 10%). The pattern of neurological status among patients with large VS versus smaller-sized VS has also been observed in other similar studies [12–15]. In other words, the majority of patients with large VS without evidence of intracranial hypertension did not differ significantly in their neurological status and symptoms when compared to patients with smaller VS.

Risk-benefit of various treatment modalities

There is no existing equipoise within the neurosurgical community regarding whether large or giant VSs should be treated. Coupled with the relative rarity of these tumors, no high-quality randomized controlled trials have compared the natural history of large VS (>3 cm) with the treatment outcomes of various modalities. Most, if not all, patients with large and giant VS undergo surgical treatment, particularly after the introduction of microsurgery in the field of neurosurgery, which has made VS excision safer. Gross total resection (GTR) or near total resection can frequently be achieved using well-established microsurgical techniques and neuromonitoring [5]. Despite this, surgical treatment continues to carry significant morbidity risks that notably increase with increasing tumor size. The risks are significantly increased to induce hearing loss (if there was serviceable hearing preoperatively) and facial nerve palsy (House-Brackmann grade III or poorer) of approximately 50% for tumors >40 mm in diameter in the setting of GTR and postoperative hematoma compared to smaller sized tumors [11]. There are other risks associated with any form of cranial surgery that are relatively small but not trivial, such as cerebellar edema, seizures, cerebrospinal fluid leakage, and infection of up to 8% overall.

Given the profoundly negative impact on quality of life (QoL) from facial nerve deficits, recent expert consensus has advocated for the goal of VS management to be shifted away from

GTR *per se*, toward “maintaining or improving QoL making every attempt at neurological function preservation with an acceptable oncological control” [16]. To this end, a planned or unplanned subtotal resection (STR) strategy to preserve facial nerve function during the surgical treatment for large and giant VS with postoperative radiation has been shown to produce excellent functional outcomes [17]. Interestingly, while tumor recurrence rates in GTR are extremely low (<10%), recurrence rates ranging from 30%–80% have been reported for patients who undergo STR of large and giant VS. Notably, several QoL studies have demonstrated a lower overall QoL for patients who have undergone surgical treatment compared to noninvasive options, including active surveillance [18]. Stand-alone upfront radiation-based options are generally not advocated for large and giant VS given the risks of swelling and edema that can produce a further mass effect in the posterior fossa. However, a more recent systemic review suggests that upfront SRS with a marginal dose of 11 Gy to 13 Gy can be offered to patients with large VS with good tumor control rates (approximately 90%) and acceptable rates of facial palsy (approximately 3%) [19].

Balancing treatment risks with natural history of disease

To adequately counsel patients on the risk-benefit of treatment options, clarity on the natural history of the disease is cardinal. In the context of neurosurgical practice, this often represents a trade-off between upfront risks of neurological morbidity that have a significant impact on the patient’s QoL versus the outlook of a disease’s impact on a patient’s neurological function over time. Existing epidemiological studies examining the natural history of VS have predominantly focused on smaller-sized VS (<30 mm). Although a very small number of large VS were included in some of these studies, there was no separate analysis of their natural history. Smaller-sized VS is generally a slow-growing entity, as approximately only one-third of the intra-meatal tumors and one-fifth of the extra-meatal increase in size within the first 5 years of diagnosis (no tumor growth occurred after the fifth year of observation), with a higher growth rate within the first year (mean of approximately 10 mm/year vs. 0.9 mm at the fourth year if growth is detected) [5]. Notably, the cutoff for the quantitative definition of growth (linear or volumetric methodologies) remains controversial, even among large extra-meatal tumors. Notwithstanding, this has shifted the treatment paradigm of smaller-sized VS toward a “watch-and-wait” approach, particularly if repeat imaging shows minimal or no growth of a newly diagnosed VS. To identify a sub-group of patients that may benefit from early treatment, several papers have also attempted to identify risk factors associated with tumor growth.

This includes features such as filling of the internal auditory canal, cystic and/or hemorrhagic content within the tumor, a history of NF2, young age, and an extra-canalicular component of greater than 20 mm. Notably, patients who were younger at diagnosis tended to have a larger VS (mean, 23 mm for the <40 years age group and mean, 13 mm for the >70 years age group). However, the association between these factors is not strong and is not widely accepted as an established risk factor independently or collectively [20].

On the other hand, a significant proportion of smaller-sized VS are known to remain static (30%–60%) in size or are even known to regress (up to 20%) since diagnosis [21]. Reddy et al. [22] retrospectively reviewed a cohort of 45 patients with medium VS (15–31 mm in diameter) over a median follow-up period of 3 years. Approximately 25% of the tumors grew, while 65% of the tumors remained static in size and just under 10% regressed in size. In other words, approximately 75% of tumors did not grow. The overall mean tumor growth was approximately 1.2 mm per year, and the growth rate among the tumors that grew varied between 2 mm and 6 mm per year. Approximately 40% of the cohort eventually underwent intervention, mostly because of tumor growth and patient choice. However, only one patient experienced “clinical worsening” necessitating surgery. The incidence of regression was noted to be double the rate for smaller VS (20%) [23,24]. The authors concluded that, for medium VS, it is reasonable to offer an initial period of “watch-and-wait” as a management option, on top of interventions.

Taken together, both small- and medium-sized VS that grow would do so in their initial phases of growth and the rate of growth is slow. Most of these tumors subsequently remain stable in size after the first few years of growth, and a sizeable proportion of them would even regress with time. The association

between tumor size at diagnosis and its propensity for growth is weak, and rarely if ever grows after a long period of non-growth or even regression [25]. Therefore, it is certainly plausible that large and giant VS share a similar tumor biology to small and medium VS; that is, the majority of tumors could be expected to remain static, with a smaller but sizeable proportion regressing after not demonstrating any significant growth during the initial phases of observation.

Over the past decade, a series of case reports have reported large and giant VS remaining static and indeed regressing over time. This occurred both in patients who had been lost to follow-up for a number of years before the observed regression on imaging, and in a couple who had been imaged annually and observed to have regressed over time. Timmer and Graa-mans [26] reported a large VS that remained static in size for >18 years and eventually started to regress toward the 22nd year of follow-up [26]. Following this, a series of case reports illustrating this phenomenon were also published. Table 1 summarizes these reports [27–30].

Plausible mechanisms

While there is much work done on the tumorigenesis of VS, the mechanisms driving tumor growth, and naturally tumor regression remain elusive [31]. Classically, the inactivation of tumor suppressor gene *NF2* located on chromosome 22q12 or the loss of its functional product moesin-erzin-radixin-like protein (*merlin*) plays a central role in tumorigenesis of VS. NF2 patients have a single germline mutation that predisposes them to VS (“double-hit” hypothesis), while other factors such as epigenetic modifications (e.g., deoxyribonucleic acid [DNA] methylation) of the *NF2* gene promoter can effectively lead to a somatic biallelic inactivation [32]. In brief, *merlin* binds to actin and stabilizes the cell membrane cytoskeleton, serves as

Table 1. Summary of case reports for spontaneous regression of large sporadic VS

Author	Year	Age at presentation (yr)	Hearing status	Size (mm)/volume (mm ³) of VS at diagnosis	Follow-up period (yr)	Size (mm)/volume (mm ³) of VS at end of follow-up period	Yearly follow-up done?	Lost to follow-up but sudden regression noted
Dahamou et al. [27]	2022	50	Right sensorineural hearing loss	37/-	3	<10/-	No	Yes, at 3-year mark
Patel et al. [28]	2021	59	Left sensorineural hearing loss	-/13,180	4	-/9,790	No	Yes, at 4-year mark
Patel et al. [28]	2021	76	No hearing loss	-/1,004	9	-/688	No	Yes, at 9-year mark
Tikka et al. [29]	2018	Not recorded	Not recorded	35/-	1	20/-	No	Yes, at 1-year mark
Tikka et al. [29]	2018	Not recorded	Not recorded	31/-	10	21/-	Yes	No
Amoo et al. [30]	2023	41	Left sensorineural hearing loss	36/-	9	21/-	Yes	No

VS, vestibular schwannomas; -, not available

a “checkpoint” for Schwann cell repair, facilitates contact inhibition of receptor-dependent promitogenic pathways, and negatively influences growth signaling pathways such as mTORC1, EGFR, and Ras/Rac. However, this genetic predisposition alone does not entirely account for tumor biology, as not all VS tumors harbor NF2 mutations. Based on single-cell ribonucleic acid (RNA) sequencing, differential expression of *merlin*-regulated genes may also result in differential pathogenicity of NF2 mutations [33]. Genes unrelated to NF2 such as HGF/c-MET; noncoding genomes (such as microRNAs and epigenetic mechanisms) have also been implicated in sporadic VS.

In recent years, there has been increasing interest in the tumor microenvironment and its influence on tumor behavior. They can be broadly classified into cellular components (immune cells, fibroblasts, platelets, etc.) and non-cellular components (structural proteins [extracellular matrix, regulatory proteins] and communication molecules [cytokines, chemokines, growth factors, etc.]), for which many entities, including macrophages, cyclo-oxygenase, and VEGF, have been identified as key players in tumorigenesis. VEGF, which plays a central role in vasculogenesis and angiogenesis, has been a target of considerable success in NF2 patients. In a meta-analysis, bevacizumab was reported to cause partial regression or change in >90% (41% regression, 47% no change in size) of patients, with approximately 10% of patients eventually requiring surgical intervention [34]. Hearing was preserved (20%–30%) or improved (50%–60%) in approximately 80% of patients. Nevertheless, serious toxicities were associated with approximately 17% of the patients, and the use of this drug may involve substantial financial costs. The off-label use of bevacizumab in sporadic VS has also been reported to produce a good response (>60% tumor volume reduction with full hearing preservation). At a macroscopic level, various studies reporting the regression of medium and large VS have also postulated tumor growth size “outpacing” the vascular supply after an initial period of growth, resulting in no growth or regression of this benign tumor. An inflammatory microenvironment has been shown to be associated with higher rates of tumor progression (increased proportion of immune cells and inflammatory cytokines such as IL-1, IL-6, and TNF) which may be driven in part by systemic inflammation [35]. It also led us to

speculate that any attempt at less than GTR could trigger a pro-inflammatory change in the tumor microenvironment post-operatively, accounting for a relatively high tumor progression rate after STR compared to GTR and even the natural growth rates of conservatively treated VS. A more recent multiplatform molecular analysis of surgically resected VS specimens using methylation and RNA profiling also revealed two robust subgroups (immunogenic and proliferative) with distinct tumor microenvironments [36]. Notably, this study noted that there were no postoperative recurrence or growth in the former group, unlike the latter group in a mean 5-year follow-up period.

Taken together, we postulate that the cause of stasis and regression of VS could be the inability of genetic and epigenetic factors driving tumorigenesis and growth to sustain a tumor microenvironment beyond its initial growth for most sporadic VS, where some function of the *merlin* gene may be variably preserved. The observation of this occurring even in large or giant VS suggests that this “balance” that influences tumor growth holds true regardless of tumor size (Fig. 3). In a significant proportion of individuals where regression is observed, changes in the local tumor microenvironment due to tumor growth may serve as a conduit or trigger a proinflammatory or apoptotic cascade that drives dramatic reductions in tumor size [37].

The phenomenon of spontaneous tumor regression has also been sparsely reported in meningiomas and gliomas [38,39]. To this end, in patients with large and giant VS with no debilitating symptoms, it might be essential to counsel them on the option of the “watch-and-wait” approach given the possibility of cessation of growth and even regression over time. Given the inherent limitations of case reports, further prospective studies are needed to better characterize the natural history of large and giant VS.

Limitations

One of the limitations of this case report is the substantial gap in follow-up period impacting the confidence in attributing the improvement in symptoms and the observed reduction in tumor size to spontaneous regression. We leveraged this opportunity to review relevant literature which make sim-

Regressive factors	Tumorigenic factors
Inadequate tumour blood supply	NF2 inactivation
Merlin gene function preservation	DNA methylation
Inflammation and apoptotic cascade	Increased HGF/c-MET expression
	Increased VEGF expression
	Increased microRNA expression
	Increased macrophage expression
	Increased COX expression

Fig. 3. Postulated balance between regressive and tumorigenic factors influencing vestibular schwannomas growth.

ilar observations and postulate mechanisms to it as well. However, analysing only case reports has its limitations. For instance, it is not possible to draw a causal link from a single uncontrolled observation. In addition, it is not possible to generalize the findings due to a lack of a representative population. This also means information regarding prevalence rates and ratios cannot be derived solely from case reports. This underlines the need for larger prospective studies including control groups which would serve to remove the likelihood of chance, bias, and confounders. Some of the details regarding possible inclusion criteria of such prospective studies are highlighted in the “Future work” section.

Future work

Regarding future prospective studies surrounding this subject, a screening protocol, based on MRI findings, needs to be included. The inclusion criteria should ensure that there is no substantial risk to the participants in having an MRI to assess volumetric growth. The authors preliminarily suggest the following criteria: patients should be aged between 18 and 70 years, patients should have been newly diagnosed with a unilateral VS less than 6 months ago, the maximal tumor diameter should be more or equal to 30 mm at the CPA and the patients should not present with any debilitating clinical symptoms and signs attributable to brainstem compression, intracranial hypertension with or without obstructive hydrocephalus, “significant” radiological brainstem compression and distortion of the fourth ventricle or with a tumor bleed. In addition, patients must also be able and willing to comply to a 3 monthly surveillance scan especially in the initial follow-up period. Patients will be excluded if there is personal history or family history of NF2 in a first degree relative.

Conclusion

We report a case of spontaneous regression of a large VS and review several similar case reports published in recent years. Taken together with similar observations in a larger series of small- and medium-sized VSs, we postulate that most large and giant VSs similarly undergo a period of growth, followed by stasis, with a significant proportion undergoing regression. This could be driven by a shift in balance that weighs away from the predominant tumorigenic factors driving tumor growth toward the predominance of regressive factors. Further studies to elucidate the natural history of large and giant VS are essential to provide adequate counseling to patients who would otherwise be conventionally offered surgical treatment that carries a significant risk of neurological morbidity.

Ethics Statement

The authors obtained informed consent from the patient to submit his case

details for this manuscript. No institutional board review approval is required to publish this manuscript based on standing institutional guidelines.

Availability of Data and Material

The authors confirm that the data supporting the findings of this study is available in the article.

ORCID iDs

Kok Ann Colin Teo  <https://orcid.org/0009-0000-8748-4009>
 Rachit Agrwal  <https://orcid.org/0009-0000-3711-6347>
 Pin Lin Kei  <https://orcid.org/0000-0003-1349-8058>
 Shiong Wen Low  <https://orcid.org/0000-0002-7506-2949>

Author Contributions

Conceptualization: Kok Ann Colin Teo, Rachit Agrwal. Data curation: Kok Ann Colin Teo. Formal analysis: Kok Ann Colin Teo, Rachit Agrwal. Funding acquisition: Kok Ann Colin Teo. Investigation: Kok Ann Colin Teo, Rachit Agrwal. Methodology: Kok Ann Colin Teo, Rachit Agrwal. Project administration: Kok Ann Colin Teo, Rachit Agrwal. Resources: Pin Lin Kei, Su Lone Lim, Siyang Ira Sun, Shiong Wen Low. Software: Kok Ann Colin Teo. Supervision: Pin Lin Kei, Su Lone Lim, Siyang Ira Sun, Shiong Wen Low. Validation: Pin Lin Kei, Su Lone Lim, Siyang Ira Sun, Shiong Wen Low. Visualization: Kok Ann Colin Teo, Rachit Agrwal. Writing—original draft: Kok Ann Colin Teo, Rachit Agrwal. Writing—review & editing: all authors.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

Funding Statement

None

Acknowledgments

None

REFERENCES

- Carlson ML, Link MJ. Vestibular schwannomas. *N Engl J Med* 2021; 384:1335–48.
- Stivaros SM, Stemmer-Rachamimov AO, Alston R, Plotkin SR, Nadol JB, Quesnel A, et al. Multiple synchronous sites of origin of vestibular schwannomas in neurofibromatosis type 2. *J Med Genet* 2015;52:557–62.
- Dewan R, Pemov A, Kim HJ, Morgan KL, Vasquez RA, Chittiboina P, et al. Evidence of polyclonality in neurofibromatosis type 2-associated multilobulated vestibular schwannomas. *Neuro Oncol* 2015;17:566–73.
- Carlson ML, Glasgow AE, Grossardt BR, Habermann EB, Link MJ. Does where you live influence how your vestibular schwannoma is managed? Examining geographical differences in vestibular schwannoma treatment across the United States. *J Neurooncol* 2016;129:269–79.
- Starnoni D, Giammattei L, Cossu G, Link MJ, Roche PH, Chacko AG, et al. Surgical management for large vestibular schwannomas: a systematic review, meta-analysis, and consensus statement on behalf of the EANS skull base section. *Acta Neurochir (Wien)* 2020;162:2595–617.
- Müller S, Arnolds J, van Oosterhout A. Decision-making of vestibular schwannoma patients. *Acta Neurochir (Wien)* 2010;152:973–84.
- Marinelli JP, Lohse CM, Grossardt BR, Lane JL, Carlson ML. Rising incidence of sporadic vestibular schwannoma: true biological shift versus simply greater detection. *Otol Neurotol* 2020;41:813–47.
- Evans DGR. Neurofibromatosis 2 [Bilateral acoustic neurofibromatosis, central neurofibromatosis, NF2, neurofibromatosis type II]. *Genet Med* 2009;11:599–610.
- Stangerup SE, Caye-Thomasen P. Epidemiology and natural history of vestibular schwannomas. *Otolaryngol Clin North Am* 2012;45:257–68.

10. Reznitsky M, Petersen MMBS, West N, Stangerup SE, Cayé-Thomasen P. Epidemiology of vestibular schwannomas—prospective 40-year data from an unselected national cohort. *Clin Epidemiol* 2019;11:981-6.
11. Samii M, Gerganov VM, Samii A. Functional outcome after complete surgical removal of giant vestibular schwannomas. *J Neurosurg* 2010;112:860-7.
12. Artz JC, Timmer FC, Mulder JJ, Cremers CW, Graamans K. Predictors of future growth of sporadic vestibular schwannomas obtained by history and radiologic assessment of the tumor. *Eur Arch Otorhinolaryngol* 2009;266:641-6.
13. Foley RW, Shirazi S, Maweni RM, Walsh K, McConn Walsh R, Javadpour M, et al. Signs and symptoms of acoustic neuroma at initial presentation: an exploratory analysis. *Cureus* 2017;9:e1846.
14. Huang X, Xu J, Xu M, Zhou LF, Zhang R, Lang L, et al. Clinical features of intracranial vestibular schwannomas. *Oncol Lett* 2013;5:57-62.
15. Gréant E, Van de Heyning P, Ihtijarevic B, Topsakal V, Menovsky T, Van Rompaey V. Sporadic vestibular schwannoma: correlation between tumour size, hearing levels, age and radiologic features in 384 patients. *B-ENT* 2020;16:97-102.
16. Nicoucar K, Momjian S, Vader JP, De Tribolet N. Surgery for large vestibular schwannomas: how patients and surgeons perceive quality of life. *J Neurosurg* 2006;105:205-12.
17. Daniel RT, Tuleasca C, George M, Pralong E, Schiappacasse L, Zeverino M, et al. Preserving normal facial nerve function and improving hearing outcome in large vestibular schwannomas with a combined approach: planned subtotal resection followed by gamma knife radiosurgery. *Acta Neurochir (Wien)* 2017;159:1197-211.
18. Carlson ML, Tveiten OV, Driscoll CL, Goplen FK, Neff BA, Pollock BE, et al. Long-term quality of life in patients with vestibular schwannoma: an international multicenter cross-sectional study comparing microsurgery, stereotactic radiosurgery, observation, and nontumor controls. *J Neurosurg* 2015;122:833-42.
19. Tuleasca C, Kotecha R, Sahgal A, de Salles A, Fariselli L, Paddock I, et al. Single-fraction radiosurgery outcomes for large vestibular schwannomas in the upfront or post-surgical setting: a systematic review and International Stereotactic Radiosurgery Society (ISRS) practice guidelines. *J Neurooncol* 2023;165:1-20.
20. Brown A, Early S, Vasilijic S, Stankovic KM. Sporadic vestibular schwannoma size and location do not correlate with the severity of hearing loss at initial presentation. *Front Oncol* 2022;12:836504.
21. Nikolopoulos TP, Fortnum H, O'Donoghue G, Baguley D. Acoustic neuroma growth: a systematic review of the evidence. *Otol Neurotol* 2010;31:478-85.
22. Reddy CE, Lewis-Jones HG, Javadpour M, Ryland I, Lesser TH. Conservative management of vestibular schwannomas of 15 to 31 mm intracranial diameter. *J Laryngol Otol* 2014;128:752-8.
23. Yamakami I, Uchino Y, Kobayashi E, Yamaura A. Conservative management, gamma-knife radiosurgery, and microsurgery for acoustic neurinomas: a systematic review of outcome and risk of three therapeutic options. *Neurol Res* 2003;25:682-90.
24. Bakkouri WE, Kania RE, Guichard JP, Lot G, Herman P, Huy PT. Conservative management of 386 cases of unilateral vestibular schwannoma: tumor growth and consequences for treatment. *J Neurosurg* 2009;110:662-9.
25. Paldor I, Chen AS, Kaye AH. Growth rate of vestibular schwannoma. *J Clin Neurosci* 2016;32:1-8.
26. Timmer FC, Graamans K. A large vestibular schwannoma that did not grow for 18 years. *B-ENT* 2011;7:305-7.
27. Dahamou M, Elfarissi MA, Lhamlili M, Mehroud I, Khoulali M, Oulali N, et al. Spontaneous regression of solid-cystic vestibular schwannoma: a case report. *Surg Neurol Int* 2022;13:414.
28. Patel EJ, Deep NL, Schecht M, Hagiwara M, Roland JT Jr. Tracking spontaneous vestibular schwannoma regression with volumetric measurements. *Laryngoscope* 2021;131:E1647-52.
29. Tikka T, Yiannakis CP, Stapleton E, Locke R, Crowther JA, Taylor WAS, et al. Spontaneous vestibular schwannoma regression: a case-control study. *Otol Neurotol* 2018;39:e1118-24.
30. Amoo M, Rawluk D, Mcconn-Walsh R, Javadpour M. The shrinking vestibular schwannoma. *Br J Neurosurg* 2023;37:701-2.
31. Tesařová M, Peterková L, Štátná M, Kolář M, Lacina L, Smetana K Jr, et al. Tumor biology and microenvironment of vestibular schwannoma-relation to tumor growth and hearing loss. *Biomedicines* 2022;11:32.
32. Pettrilli AM, Fernández-Valle C. Role of Merlin/NF2 inactivation in tumor biology. *Oncogene* 2016;35:537-48.
33. Xu M, Wang S, Jiang Y, Wang J, Xiong Y, Dong W, et al. Single-cell RNA-Seq reveals heterogeneity of cell communications between Schwann cells and fibroblasts within vestibular schwannoma microenvironment. *Am J Pathol* 2022;192:1230-49.
34. Lu VM, Ravindran K, Graffeo CS, Perry A, Van Gompel JJ, Daniels DJ, et al. Efficacy and safety of bevacizumab for vestibular schwannoma in neurofibromatosis type 2: a systematic review and meta-analysis of treatment outcomes. *J Neurooncol* 2019;144:239-48.
35. Hannan CJ, Lewis D, O'Leary C, Donofrio CA, Evans DG, Roncaroli F, et al. The inflammatory microenvironment in vestibular schwannoma. *Neurooncol Adv* 2020;2:vdaa023.
36. Landry AP, Wang JZ, Suppiah S, Zadeh G. Multiplatform molecular analysis of vestibular schwannoma reveals two robust subgroups with distinct microenvironment. *J Neurooncol* 2023;161:491-9.
37. Amit M, Xie T, Gleber-Netto FO, Hunt PJ, Mehta GU, Bell D, et al. Distinct immune signature predicts progression of vestibular schwannoma and unveils a possible viral etiology. *J Exp Clin Cancer Res* 2022;41:292.
38. Kim BS, Jung TY, Moon KS, Kim IY, Jung S. Meningioma with partial and spontaneous regression of peritumoral edema on long-term follow up. *Brain Tumor Res Treat* 2022;10:275-8.
39. Parsa CF, Hoyt CS, Lesser RL, Weinstein JM, Strother CM, Muci-Mendoza R, et al. Spontaneous regression of optic gliomas: thirteen cases documented by serial neuroimaging. *Arch Ophthalmol* 2001;119:516-29.